

FR FRANCE RAG ASSISTANT

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1. CHUNKING STRATEGY

Strategy: Fixed-size overlapping chunks (100 words, 20 overlap)

- Prevents breaking context across chunks**
- Keeps each chunk answerable independently**
- Overlap helps retain sentence continuity**

Why 100? Best trade-off between granularity and context window.

Why overlap? It mitigates cutoff errors during embedding windowing.

2. EMBEDDING MODEL SELECTION

Chosen: all-MiniLM-L6-v2

- **Dim: 384 | Speed: Fast | Accuracy: Strong baseline**
- **Compatible with FAISS FlatIP for real-time search**

Compared to: m2-bert-80M-8k-retrieval

- **Pros: Larger context window**
- **Cons: Slower, larger index, GPU-intensive**

Trade-off: MiniLM is ideal for low-latency semantic search.

3. HYBRID RETRIEVAL DESIGN

Base: Cosine similarity with FAISS over dense vectors

Filter: Metadata-aware filtering by category (e.g., 'Geography')

Example:

Query: 'What are the mountains in France?'

→ Filtered to 'Mountains', 'Alps', 'Massif Central'

→ Improves precision by discarding unrelated soil/climate chunks

4. PROMPT EVOLUTION & DESIGN

Initial Prompt: Open-ended LLM use → hallucinations observed

Final Prompt:

"You are a helpful assistant. Use ONLY the provided context. If unsure, say 'I don't know'."

Benefits:

- Prevents over-generation**
- Keeps answers grounded in retrieved facts**
- Enables traceability & safety**

5. PERFORMANCE HIGHLIGHTS

Strong Case:

Query: 'What are the major mountain ranges in France?'

→ Chunks: Alps, Massif Central, Hercynian → Clear, accurate answer

Weak Case:

Query: 'Why is France's soil so fertile?'

→ Chunks mixed: climate/soil not aligned → vague LLM output

Insight: LLM depends heavily on top-k chunk relevance

6. FUTURE WORK

Improve Reranking:

- Use reranker LLM after top-k retrieval to boost precision**

Language Support:

- Add multilingual chunking & query understanding**

Vector Store Upgrade:

- Replace FAISS with persistent ChromaDB or Qdrant**

Feedback Loop:

- Store user ratings and fine-tune top-k/LLM prompts dynamically**